

Topic: Analysis of TCP's Congestion Control using ns-3

CS 3205: Assignment 3

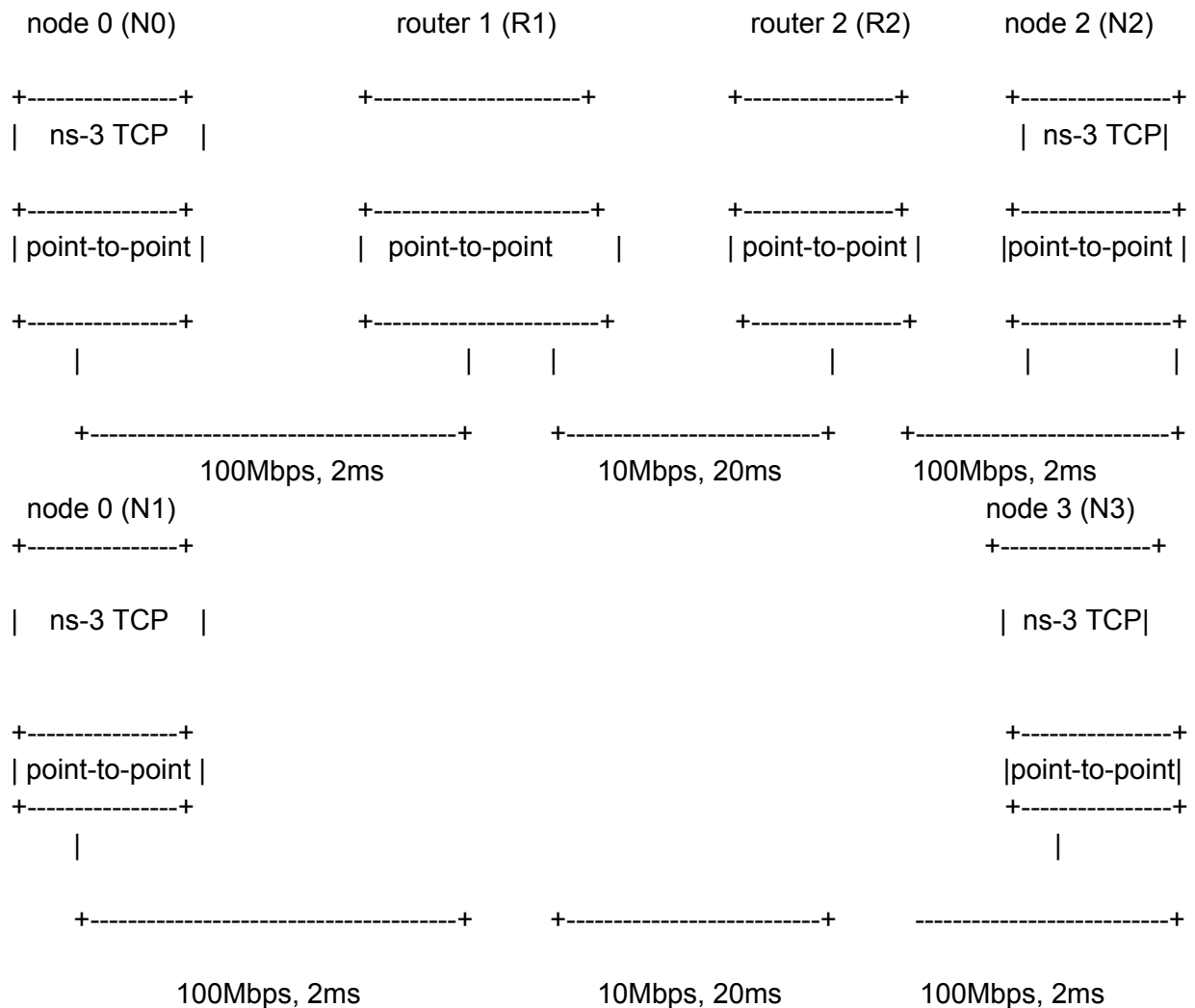
The ns3 script for three nodes, tcp-example-cwnd.cc, is attached.

This script simulates a network of three nodes (say, N0 and N1, N2) with following details.

Bandwidth between N0 & N1 = 5Mbps; delay = 5ms

Bandwidth between N1 & N2 = 5Mbps; delay = 5ms

Data is transferred over TCP on the link using the BulkSend Application. Modify this script to the following topology



Modify it to **n0, n1** -> sender nodes, **n2, n3** -> receiver nodes, **r1, r2** -> routers

The link between **r1** and **r2** acts as the **bottleneck**

Link Configuration

Link	Bandwidth	Delay
Sender <-> Router	100 Mbps	2 ms
Router <-> Router (bottleneck)	10 Mbps	20 ms
Router <-> Receiver	100 Mbps	2 ms

Use **TCP applications** BulkSendApplication and PacketSink to generate traffic.

Use **FlowMonitor** to collect per-flow statistics such as throughput and received bytes.

Parameter	Default value
TCP variant used	TcpNewReno/TCP Cubic
Simulation time	10 seconds
Application payload size	1460 bytes

Upon running successfully, the "tcp-example-cwnd.cc" script produces the following output files.

- pcap logs collected at nodes N0, N1 and N2 The congestion window of the TCP sender at node N0 is recorded in "tcp-example.cwnd".
- "tcp-example.tr" is the ns-3 ASCII trace file containing all simulator events. The ns-3 documentation should explain this trace file in detail.

Analyzing simulation results

For your assignment, you will need to run simulations by creating the topology specified. Analyze the output files to answer questions about TCP. To understand what is happening with TCP in the simulation, it will be helpful to look at the following metrics.

- **Average TCP throughput at the receiver.**
Flowmonitor provides the average throughput, in addition it can be easily obtained by opening the receiver's pcap file in wireshark, and checking out the "Summary" tab or "Conversations" tab under the "Statistics" menu. (For calculation of average throughput, counting the bytes in TCP headers is optional - either way shouldn't matter much.)
Alternatively, you can write a script to analyze the pcap file and calculate throughput. You can use "tshark -r" or "tcpdump -r" with several useful options to read the pcap files and process them. Please read up online about the tshark tool and its various options. For example, if you run tshark with the "-T fields" option, you can extract only a subset of

fields from the pcap file to the terminal output, which you may then redirect to a smaller file that is easy to parse. For example, the following command will read the pcap file "tcp-example-0-0.pcap" and extract the timestamp, source IP, destination IP, port numbers, and sequence numbers.

```
$tshark -r tcp-example-0-0.pcap -T fields -e frame.time_epoch -e ip.src -e ip.dst -e tcp.port -e tcp.seq -e tcp.len -e tcp.ack
```

You can use several such simple tshark commands to easily extract only the information you care about from the pcap files. After you extract information from the pcap file, you may use sed/awk/perl/python or even C++/Java to extract the various columns in the output, and perform any manipulations (e.g., add up the packet length received column and divide by total time to get throughput). You can compute throughput for a window say for example add up the packet length column for a window of 2 sec, divide the number of bytes by 2 sec to compute throughput for that 2 sec.

- **Evolution of the sender's congestion window (cwnd) over time.** The cwnd trace file, generated as part of the simulation output, is written whenever cwnd changes. So simply plotting all the values in the cwnd trace file is enough for you to visualize what happened to the cwnd during the simulation. You may use any graph plotting software. For example, here is a simple gnuplot script, say, "example.gpp", that generates a cwnd vs time plot.

```
set term postscript eps color  
set output 'cwnd.eps'  
set ylabel 'cwnd'  
set xlabel 'time'  
plot 'tcp-example.cwnd' using 1:2
```

Store the 5 lines above in a file called "example.gp", and store the cwnd trace file generated by ns-3 called "tcp-example.cwnd" also in the same folder. Then run "gnuplot example.gp" at the command line. The graph "cwnd.eps" will be created in the same folder showing the cwnd vs time plot.

In general, once you generate the output data to plot, a simple variant of a gnuplot script like this will let you generate any graph you want.

Create a report to write the answers for the following questions.

Q.1. Run the simulation for 10sec and answer the following questions. Create one TCP flow (n0 -> n2).

- a. What is the maximum expected value (theoretical) of throughput (in Mbps)? Why? [1+1]
- b. How much is Bandwidth-Delay-Product (BDP)? Express your answer in terms of the number of packets. [1]

- c. What is the average computed throughput of the TCP transfer from simulation using flow-monitor? Plot throughput with time from pcap (compute throughput for each 2 sec and plot that with time) [2]
- d. Is the achieved throughput approximately equal to the maximum expected value? If it is not, explain the reason for the difference. [1+1]
- e. Plot Congestion Window (CWND) with time [1]

Q.2 Two Competing Flows, Add a second flow (n1 -> n3). Both share the bottleneck. Measure the throughput of each flow. [1]

- a. Are the flows getting equal bandwidth? [1]
- b. Why does TCP tend toward fair sharing? [1]

Q3. Multiple Parallel Connections: create multiple flows at n1

- a. User 1: 3 TCP flows from n0->n2 [1]
- b. User 2: 1 TCP flow [1]
- c. Does User 1 get more bandwidth? [1]
- d. Does TCP provide per-flow or per-user fairness? [1]

Q4. TCP Variant Comparison: Repeat Q2 using two TCP variants (e.g., NewReno, CUBIC).

- a. Do variants behave differently? Compare their CWND using a plot. [2]
- b. Which is more aggressive? [1]

Deliverables:

Submit a zip/tar file containing your source code for simulation, scripts you used for analysis and plotting, and a report of the observations.